Brody Stephano

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OBJECTIVE

Motivated Electrical Engineering undergraduate with hands-on experience in analog and RF systems, seeking to advance into a master's program in analog integrated circuit design.

EDUCATION

Florida Atlantic University Boca Raton, FL

Bachelor of Science in *Electrical Engineering* Expected Spring 2026

GPA: 3.8

TECHNICAL SKILLS

Hardware: Oscilloscopes, Multimeters, Spectrum Analyzers, Smith Chart

Software: Advanced design systems, Altium, KiCad, AutoCAD, Revit, C, C++, Arduino, P-Spice, MATLAB

RELEVANT COURSEWORK

- Microwave Engineering, High Frequency Amplifier Design, RF Devices & Circuits, Electronics 2, Electronics 1, Power Electronics, Principles of Communication Systems, Electromagnetic Fields and Waves

PROJECTS

DSBSC Transmitter

- 17 discrete-transistor RF signal chain implementing DSB-SC modulation using Gilbert cell mixer topology. Designed dual cross-coupled LC oscillators to generate carrier (1.07 MHz) and baseband (41 kHz) signals, along with their respective buffers to ensure minimization of oscillator, mixer, and mixer output feedthrough. Applied nonlinear circuit analysis and biasing techniques to ensure proper mixer operation and signal multiplication. Simulated full system in ADS, verifying spectral output and modulation behavior.

Voltage Controlled Cross-Coupled LC Oscillator

- 5 discrete-transistor RF signal chain, with a tuning range of 100 kHz. Designed across-coupled LC oscillator to generate carrier at 3 MHz, and a Colpitts oscillator for the baseband signal at 15 kHz. Simulated full system in ADS, verifying spectral output and oscillator behavior.

BJT Audio Amplifier

- Designed and developed a four-stage BJT audio amplifier featuring a cascode differential input stage with Wilson current mirror biasing, a common-emitter voltage gain stage, and a power amplifier with voltage feedback for enhanced linearity. Achieved a linear voltage gain of 90 across an 8 Ω load with a 30 kHz bandwidth.

Colpitts Oscillator

- Designed and analyzed a high-frequency LC Colpitts oscillator using a BJT. Conducted startup and small-signal analysis, verified oscillation conditions, and optimized bias network for clean sine wave output.

Heart PCB

- Designed a custom heart-shaped PCB for my girlfriend using a 555 timer-based LED circuit using RC characteristics to control variable pulse rate. Designed schematic and layout in Altium for manufacturing.

EXPERIENCE

- **Pixels On Target — Electrical Engineering Intern *(Full-Time Offer for May 2026)***
  December 2025 – Present
  Designed and analyzed analog and digital video signal chains with focus on filtering, gain, and noise performance; evaluated reliability across wide temperature ranges. Performed EMI testing, reviewed high-frequency PCB layouts in Altium, and supported validation and debugging of mixed-signal systems.
- **Thompson and Youngross Engineering Consultants**, May 2024 - January 2025, May 2025 – August 2025
  Created AutoCAD drawings for commercial and residential applications, checked design compliance, and performed load balancing using NEC/NFPA standards.

ACHIEVEMENTS

- FAU President's List Fall 2024 recipient.
- Dean's List Spring 2023, Fall 2023, Spring 2024, Summer 2024, Spring 2025, Fall 2025 recipient.